

Week 7 – Prompt Engineering for Business Intelligence

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Section 1 – Final Edited Board-Level KPI Summary

The Business Intelligence dashboard for HimShakti's Food Processing Raw Material Expiry & Inventory Waste Management system provides a comprehensive overview of inventory performance and operational efficiency. The dashboard tracks key performance indicators such as Total Inventory Quantity, Remaining Inventory Quantity, Monthly Batch Entries, Expiring Soon Batches, Average Shelf Life, Inventory Utilization Rate, and Inventory Waste Percentage. These metrics help management monitor inventory status, identify potential expiry risks, and support data-driven production planning.

The analysis indicates that inventory levels and procurement activities remain stable across the recorded period, while monitoring expiring batches enables timely stock utilization and minimizes raw material wastage. The dashboard also highlights the relationship between inventory quantity and remaining stock, allowing warehouse managers to identify excess inventory and improve stock rotation practices. Average shelf life analysis supports better production scheduling by helping prioritize raw materials that require immediate processing.

From a business perspective, the dashboard enables proactive inventory management rather than reactive decision-making. Management should regularly review KPI trends, monitor batches approaching expiry, and maintain accurate inventory records to reduce operational losses. Implementing automated expiry alerts, improving procurement planning, and continuously updating inventory data will further strengthen inventory control and operational efficiency. Although the current dashboard provides

valuable insights, future improvements such as supplier performance analysis, demand forecasting, and predictive inventory analytics can enhance long-term business planning. Overall, the dashboard supports informed decision-making by transforming inventory data into meaningful business insights while promoting efficient resource utilization and minimizing inventory waste.

Section 2 – Changes Log

S. No.	AI Generated Statement	Edited Statement	Reason for Edit
1	Inventory waste has increased significantly over time.	The dashboard identifies inventory waste through expiring batches, but the available data does not confirm an increasing trend over time.	The original statement assumed a trend that was not supported by the dataset.
2	All batches with high remaining quantity indicate overstocking.	High remaining inventory only represents available stock. Overstocking can only be confirmed after analysing demand and consumption patterns.	Additional business data is required before concluding overstocking.
3	Monthly batch entries directly represent customer demand.	Monthly batch entries represent procurement activity and inventory inflow, not actual customer demand.	Procurement and customer demand are different business metrics.
4	Products with longer shelf life always generate higher profits.	Shelf life improves storage flexibility, but profitability depends on production cost, sales volume, and market demand.	Profitability cannot be determined using shelf life alone.

5	The dashboard predicts future inventory shortage	The current dashboard monitors existing inventory performance. Predictive forecasting would require additional historical data and forecasting models.	The current dashboard is descriptive, not predictive.
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